

Final Report: Geopolitical Conflict Risk Dataset

Problem Statement

As instability and regional conflict appear to be increasing globally, this project asks a single question: which countries are most likely to experience conflict in the next six months, based on a multi-factor assessment of their month-to-month geopolitical conditions? To answer it, I built a predictive model trained on Kaggle's Global Geopolitical Conflict Dataset (2020–2025), which compiles monthly indicators commonly used to gauge regional instability—inflationary pressures, military budgets, income disparity ratios, and press freedom scores, among other economic and event-based factors. The model learns from historical escalations using the conditions present six months prior to each event, so that current data can be fed in to flag the countries at greatest near-term risk. Success is defined by two goals: maximizing predictive accuracy on historical data while guarding against overfitting, and producing actionable, country-level risk assessments. The approach is bounded by real constraints—diplomatic efforts that defuse tensions are often invisible in the data, and composite measures like press freedom scores can be imprecise—so the model's output is best understood as a risk signal to inform human judgment rather than a deterministic forecast.

Data Wrangling

The initial Kaggle dataset came to me relatively clean, but with a few idiosyncrasies that would require delicate treatment. Initially, the dataset featured 25 columns with 1320 non-null values each with datatypes ranging from continuous to categorical. I began by familiarizing myself with the various columns and their meanings within the context of the data and then assessing unique values and datatypes to determine which values were categorical strings, categorical numbers and which values were strictly numeric.

The lack of ANY null values immediately raised concerns for me and it became immediately apparent that this was largely due to bad data. Certain columns like the election cycle column, had all zeros for all 1320 rows. I elected to drop these features. Then, identified an issue with the “last_conflict_year” column which was a monthly value defining the last year which a country had experienced conflict. I determined this might be useful for future analysis and quickly cleaned this data by applying the datetime object and using boolean masks to filter identify and replace bad values.

I then setup filters for identifying potential outliers in the dataset, visualized distributions and began EDA.

Exploratory Data Analysis

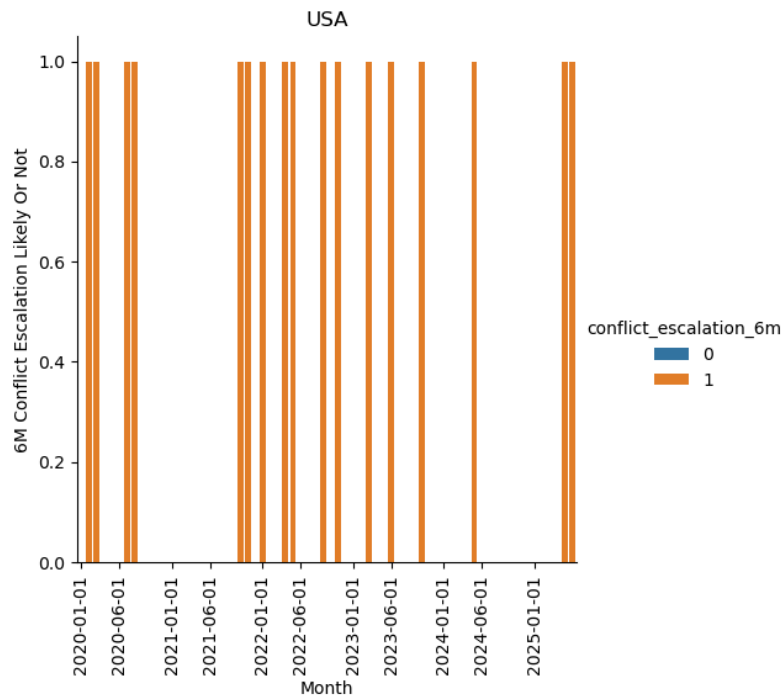
Immediately, I identified this dataset as a strong candidate for a multivariate logistic regression. I began by building correlation matrixes and assessing the relationship between continuous variables and the target variable “6m_conflict_escalation”. Many multicollinear variables

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presented themselves, helping me to trim certain model features and reduce dimensionality while lending valuable insights about how features might interact with the target value.

Because this was a time series dataset grouped by numerous countries, I then decided to investigate patterns as time series features. I produced country-by-country visualizations of the observed time period for the data and mapped the “conflict_escalation_6m” on the timeline.

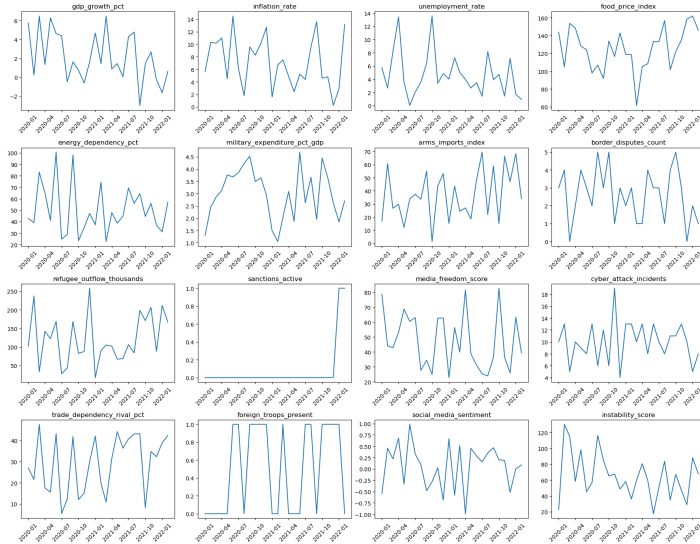


*Example of one chart for the USA (mapped all countries).

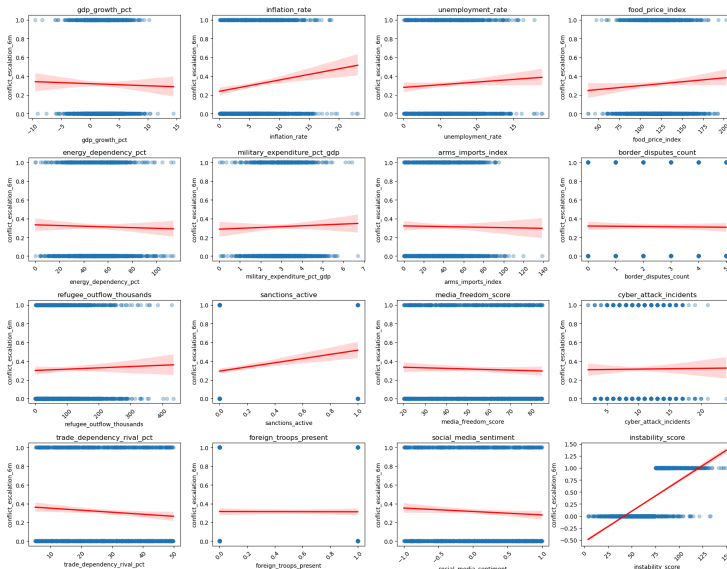
I was immediately surprised to notice that the “conflict_escalation_6m” variable didn’t seem to accurately map to real-life conflicts in many cases, for instance, no alarms went off predicting the USA’s Jan 6 Insurrection. Leading to doubts about the validity of the data at all or the definition of a conflict escalation.

I proceeded to map the variables as time series country-by-country and juxtapose those values against the target variable in hopes of identifying visual relationships.

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I spent a little bit of time exploring different countries that I knew had experienced different forms of conflict during the observation time period, such as Russia and Ukraine vs. civil conflict. And certain patterns began to emerge, like arms imports and military spending increasing in the months before an interstate armed conflict.



I then plotted a regression plot across these features to help visualize strong relationships. Confirming many of my initial suspicions and identifying instability score as a near perfect correlation - which may actually have been a confounding variable.

Modeling

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When it came to pre-processing and modeling, I prepared a time series based training/test split of the data since where data was trained on past values and expected to predict future outcomes from the datasets.

Feature engineering included adding lags and rolling averages to some continuous features i expected to see a relationship with following EDA and getting dummies for categorical features. This quickly ballooned my datasets width, so i elected to include dimension reduction in my pipeline.

I built a pipeline which standardized and scaled continuous values, then applied PCA and constructed a Logistic Regression model from the PCA features. This would become my baseline model and it would show moderate initial success, although i was disappointed by the class-1 recall being only marginally better than a coin flip.

	precision	recall	f1-score	support
0	0.86	0.92	0.89	250
1	0.78	0.65	0.71	110
accuracy			0.84	360
macro avg	0.82	0.79	0.80	360
weighted avg	0.84	0.84	0.83	360

To solve for the difficulty calling positives accurately, I applied a gradient boosting model through LightGBM which demonstrated an immediate an power improvement over baseline.

0	0.98	0.92	0.95	250
1	0.84	0.95	0.89	110
accuracy			0.93	360
macro avg	0.91	0.94	0.92	360
weighted avg	0.94	0.93	0.93	360

Then applied CVGridSearch to tune hyperparameter features, ultimately finding a model that was a strong fit for the dataset.

	precision	recall	f1-score	support
0	0.99	0.91	0.95	250
1	0.83	0.97	0.90	110
accuracy			0.93	360
macro avg	0.91	0.94	0.92	360
weighted avg	0.94	0.93	0.93	360

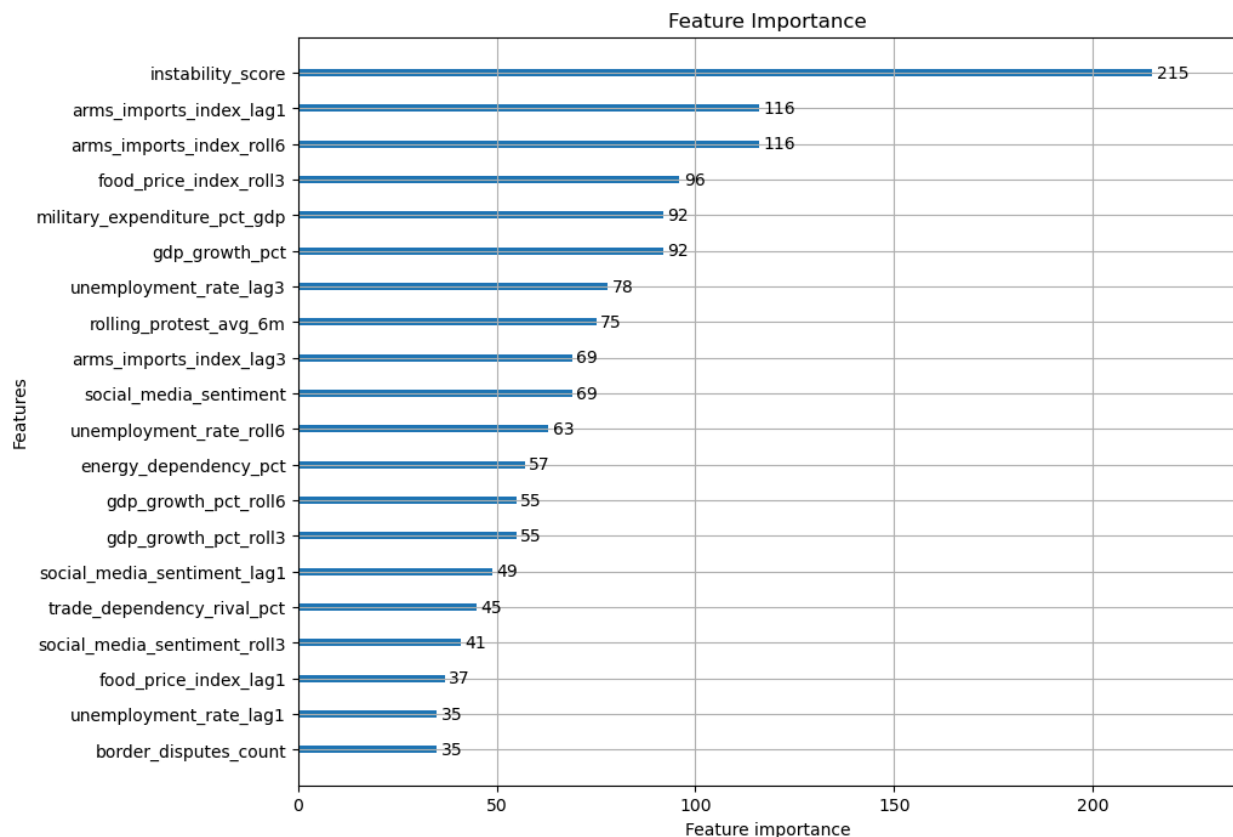
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I attempted one final model in the form of a Support Vector Machine (SVM) and attempted hyperparameter tuning, but could not deliver a superior result to the LightGBM model.

In-Depth Analysis

While the model's accuracy itself achieved the goals of this study in developing a system for predicting when conflict escalations might occur, the features that ended up bubbling to the surface as a result of this model were particularly interesting.



Instability score was the strongest feature in predicting whether or not a conflict_escalation_6m alert would trigger, but interestingly lag1 arm_imports and 6m rolling arms imports would be the 2nd and 3rd features which best determined whether or not an escalation would occur. This finding makes sense given the material covered by the dataset and heavily implies that the 6m conflict escalation feature was examining armed interstate conflicts rather than internal civil conflicts. Or at the very least that it was easier to see the writing on the wall in those situations.

Conclusion

This project began with a straightforward question which countries are most likely to experience a conflict escalation in the next six months and ended with a model that answers it well on paper: a hyperparameter-tuned LightGBM classifier that reached 93% accuracy and, more importantly, caught 97% of actual escalations in the test set, a dramatic improvement over the PCA and logistic regression baseline that could barely beat a coin flip on the positive class. Just as valuable as the accuracy, though, were the features that rose to the top. Instability score was

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the dominant predictor, though its near-perfect correlation makes me suspect it functions as a confounder rather than a true leading signal. The more telling result was that lagged and six-month rolling arms imports came in second and third, a pattern that strongly implies the [conflict_escalation_6m](#) target is capturing interstate armed conflict, where military buildup is visible months ahead, rather than internal civil unrest, which the model and the data consistently failed to anticipate (the absence of any alarm before January 6 being the clearest example). That limitation is the real takeaway. The model learned the dataset's labeling logic faithfully, but that logic is an imperfect and sometimes erratic proxy for real-world conflict, so the output is best treated as a risk signal to direct human attention rather than a forecast to act on directly. With a more rigorously defined target and broader, better-validated data, this framework could become genuinely useful—but the harder lesson is how much of conflict, especially the diplomatic and civil kind, remains invisible to the indicators we currently know how to measure.